

Object-Oriented Programming (OOP) is a programming paradigm based on the concept of objects, which contain data and code. The four main pillars of OOP are encapsulation, inheritance, polymorphism, and abstraction.

Encapsulation means bundling data and the methods that operate on that data within a single unit, usually a class, and restricting direct access to some of an object's components.

Inheritance allows a class to inherit properties and behavior from another class, promoting code reuse. The class that inherits is called a subclass, and the class being inherited from is called a superclass.

Polymorphism allows objects of different classes to be treated as objects of a common superclass, enabling one interface to represent different underlying forms.

Abstraction means hiding complex implementation details and showing only the necessary features of an object.